MMCS Report

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Abstract

This report presents a strategic model for optimizing scheduling and advertising strategies in the television broadcasting sector, aimed at maximizing viewership and advertising revenue. Utilizing a simulated dataset representing a competitive environment of four TV channels, the model focuses on intelligent scheduling and cost-effective advertising to enhance audience retention and expansion. The methodology leverages stochastic modeling to predict viewership dynamics and advertising effectiveness, accommodating real-world complexities such as varying viewer demographics and competitive actions. This approach enables a data-driven strategy that systematically adapts to competitors' schedules, targeting optimal times for movie broadcasts and advertisement placements. Key challenges include the integration of diverse data features, such as movie popularity, viewer demographics, and ad slot pricing, to formulate a robust operational strategy that aligns with budget constraints and revenue goals.

Contents

1 Literature Review **1**

2 Introduction **1**

3 Model Formulation **2**
3.1 General Sets and Notations . . . 2
3.2 Upper Layer Model . . . 2
3.2.1 Parameters . . . 2
3.2.2 Variables . . . 3
3.2.3 Constraints . . . 4
3.2.4 Objective Function . . . 4
3.3 Lower Layer Model . . . 5
3.3.1 Specific Sets and Notations . . . 5
3.3.2 Parameters . . . 5
3.3.3 Variables . . . 5
3.3.4 Constraints . . . 6
3.3.5 Objective Function . . . 7

4 Results and Analysis **7**
4.1 Analysis on Viewership and Profit . . . 7
4.2 Time Series Analysis . . . 8
4.3 Advertisement Cost Analysis . . . 9
4.4 Revenue Analysis . . . 9
4.5 Demographic Analysis . . . 10

5 Conclusion **10**

6 Annex **11**

1 Literature Review

Over the past decade, the television industry has undergone profound changes that have redefined its global operations. Once-dominant networks that controlled a significant portion of the prime-time viewing audience have seen their share steadily decline as viewers increasingly turn to alternative platforms. This shift in consumer behavior has been fueled by the rise of streaming services, which offer a vast array of on-demand content, often rendering traditional cable and satellite TV less relevant. At the same time, the cost of producing high-quality programming has surged, with an hour of prime-time content now costing upwards of 1 million USD [4]. This dramatic increase in production costs, coupled with growing competition from streaming platforms and other digital media outlets, has contributed to a noticeable decline in the profitability of traditional TV programming.

The internet, in particular, has played a central role in this transformation, marking a shift from conventional broadcast models to more flexible, personalized viewing experiences. Platforms like Netflix, Amazon Prime, and Disney+ have revolutionized how audiences consume television, offering a wealth of content that can be accessed on-demand, at any time, and from virtually anywhere [1]. This shift highlights the importance for traditional television businesses to adapt and optimize their resources in order to remain competitive and maximize profitability in an increasingly fractured market. To stay relevant and profitable, TV channels must leverage their unique assets while exploring new business models that align with the evolving landscape of digital entertainment [5].

2 Introduction

This report focuses on optimizing scheduling and advertising strategies for a television broadcasting channel, with the goal of maximizing viewership and advertising revenue in a highly competitive environment. The primary objective is to develop a strategic model that enhances movie scheduling and advertising slot allocation. By analyzing simulated scheduling data from the channel and its three main competitors, the model aims to improve audience retention, expand the viewership base, and ultimately boost advertising revenue. This report seeks to identify effective strategies that balance programming choices with advertising placements to achieve optimal performance in a competitive broadcasting market.

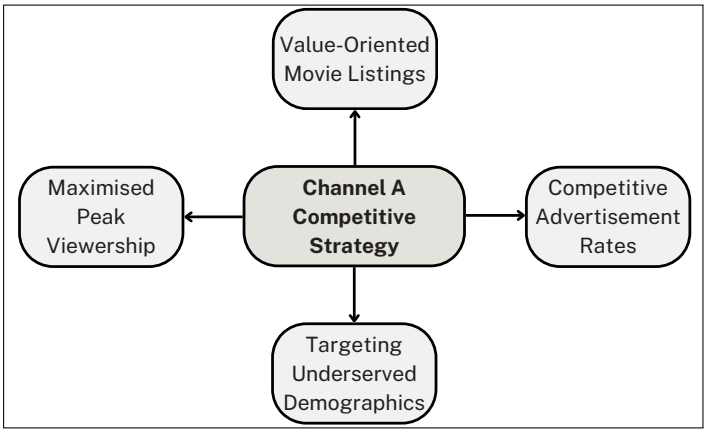


Figure 1: Channel A Strategy

This infographic illustrates Channel A's strategic approach to maintaining competitiveness across multiple areas:

- **Competitive Advertisement Rates**: Channel A offers advertising slots at competitive rates, dynamically adjusting prices based on viewership. By benchmarking against market prices, the strategy ensures fairness and encourages advertiser investment. The pricing of advertisement slots is determined by incorporating a viewership factor. As viewership increases during a particular time slot, the corresponding advertisement rate is adjusted upward, reflecting its increased value.
- **Value-Oriented Movie Listings**: Movies are selected based on a cost-effectiveness ratio of popularity to license fee, ensuring high-value content with minimal cost.
- **Maximized Peak Viewership**: Content is scheduled to capture peak viewership during periods of low competitor activity, leveraging time series analysis to identify market opportunities.
- **Targeting Underserved Demographics**: Channel A targets underserved audiences, such as retirees, by scheduling content that aligns with their habits and featuring older movies that appeal to them.

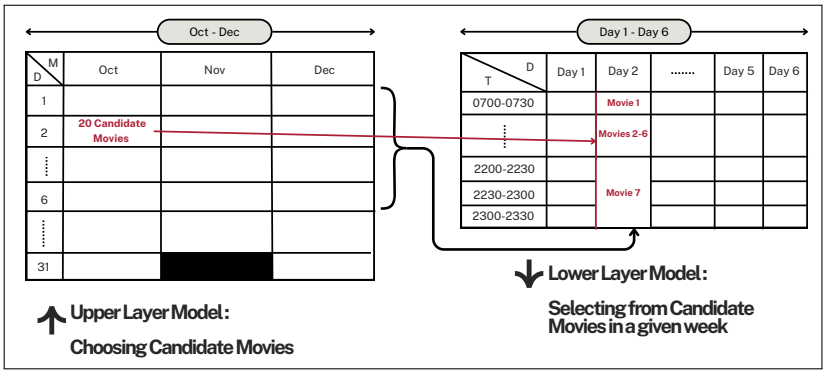


Figure 2: Upper layer to lower layer model

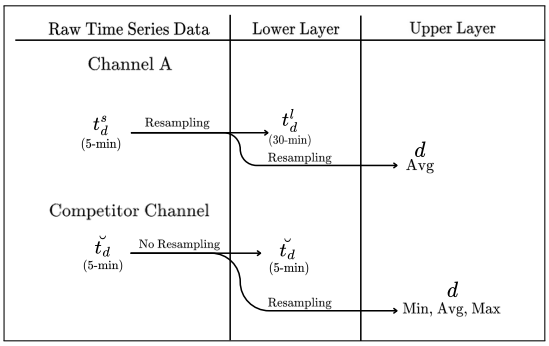


Figure 3: Changes in time series data

Two-Layer Model Approach (Figure 2)

- Our approach is a two-layer model [3], with the upper layer identifying candidate movies that offer good value for money and allocating both movies and advertisements for each day over a three-month period.
- The lower layer then schedules the movies and advertisements for each specific day. This layered approach is designed to reduce the runtime required to obtain an optimal solution.

The preprocessing methodologies for raw time-series data, initially recorded at five-minute intervals, are described for each respective model. As shown in Fig.3 ,The upper layer model, which chooses candidate movies across a three-month horizon at daily resolution, requires the resampling of all data to a one-day timeframe. This was achieved by computing the mean, minimum, and maximum values. For the lower layer model, the data from Channel A was resampled to a thirty-minute timeframe using mean values, while the data from the competitor channel was maintained at its original five-minute resolution.

3 Model Formulation

3.1 General Sets and Notations

- D = Set of days = $\{0, 1, \ldots, 83\}$
- W = Set of weeks with respect to days = $\{w_0 = \{0, 1, \ldots, 5\}, w_1 = \{6, 8, \ldots, 12\}, \ldots, w_{12} = \{83\}\}$
- Φ = Set of Months with respect to weeks = $\{\phi_0 = \{w_0, \ldots, w_4\}, \phi_1 = \{w_4, \ldots, w_8\}, \phi_2 = \{w_4, \ldots, w_{12}\}\}$
- M = Set of Candidate Movies, $M = \{1, 2, 3, \ldots, 5919\}$
- C = Set of Competitor Channels = $\{0, 1, 2\}$
- G = Set of Movie Genre, G = {"horror", "sci-fi", "romance", ...}
- P = Population Demographic Set, $P = \{"children", "adult", "retirees"\}$
- T^l = Thirty-minute timeframe slot for a single day time horizon, $T^l = \{0, 1, 2, \ldots, 33\}$
- T^l_d = Thirty-minute timeframe slot for each day d, $T^l_d = \{34d, 34d + 1, \cdots, 34d + 33\}; d \in D$
- T^s = Five-minute timeframe slot for a single day time horizon, $T^s = \{0, 1, 2, \ldots, 203\}$
- T^s_d = Five-minute timeframe slot for each day, $T^s_d = \{204d, 204d + 1, \cdots, 204d + 203\}; d \in D$
- $\breve{T}_d$ = Competitor's advertisement time slot , $\breve{T}_d \subseteq T^s_d$

3.2 Upper Layer Model

3.2.1 Parameters

- $R^{ads_s}_m$: Five-minute Runtime slot with advertisement of movie m. $(m \in M)$
- F_t : Prime time factor at time t. $(t \in T^s_d)$
- $B_{p,t}$: Baseline Viewership of demographic p at time t. $(p \in P, t \in T^s_d)$
- $P_{m,p}$: Popularity of movie m for demographic p $(m \in M, p \in P)$

- $License_m$ License fee for each movie m. We note that the license fee of the movie is dependent on the movie budget and movie revenue. $(m \in M)$
- N_m^{ads}: Number of advertisements being played during movie m. $(m \in M)$
- $N_{c,d}^{ads}$: Number of advertisements that opponent channel c shows on day d $(c \in C, d \in D)$
- $Z_{m,t}$: Expected viewership with prime time factor for at time t of movie m.$(t \in T_d^s, m \in M)$

$$Z_{m,t} = P_{m,p} B_{p,t} F_t.$$

- $Z_{m,d}^{Avg}$: Average Expected baseline viewership with prime time factor for day d of movie m . $(d \in D, m \in M)$
- $Cost_{c,\breve{t}}^{Ads}$: Cost of buying an advertisement slot from channel c on at time $\breve{t}$. $(c \in C, \breve{t} \in \breve{T}_d)$
- $Cost_{c,d}^{ads}$: Average cost of buying an advertisement slot from channel c on day d. $(c \in C, d \in D)$
- $Rev_{m,t}$: Revenue of selling an advertisement slot during movie m at time t is determined by the original ad slot price multiplied by the popularity factor,$(m \in M, t \in T_d^s)$,

$$Rev_{m,t} = \frac{Z_{m,t} \cdot N_m^{ads}}{License_m} \cdot \frac{Z_{m,t}}{\min_t\{Z_{m,t}\}}.$$

- $Rev_{m,d}^{Avg}$: Average revenue of selling an advertisement slot during movie m on day d. $(d \in D, m \in M)$
- $Conv_{c,\breve{t}}$: Conversion Rate if we show advertisements at time $\breve{t}$ on channel c.$(\breve{t} \in \breve{T}_d, c \in C)$
- $TVC_{c,\breve{t}}$: True viewership count at time $\breve{t}$ on channel c $(\breve{t} \in \breve{T}_d, c \in C)$
- $U_{m,g}$: Binary matrix indicating if movie m shows genre g. $(g \in G, m \in M)$

$$U_{m,g} = \begin{cases} 1 & \text{if the move } m \text{ is shows genre } g; \\ 0 & \text{otherwise.} \end{cases}$$

- $I_{c,m,d}$: Binary matrix indicating if movie m is played on channel c on day d. $(c \in C, m \in M, d \in D)$

$$I_{c,m,d} = \begin{cases} 1 & \text{if the channel } c \text{ is shows movie } m \text{ on day } d; \\ 0 & \text{otherwise.} \end{cases}$$

- $W_{m,d}^{Avg}$[Million Views/£] = Average value of choosing movie m as a candidate movie on day d. $(m \in M, d \in D)$

$$W_{m,d}^{Avg} = \frac{1}{204} \sum_{t=204d}^{204d+203} \frac{Z_{m,t} \cdot N_m^{Ads}}{License_m}.$$

- $A_{c,m,d}^{Avg}$[Million Views/£] = Average value of playing advertisement of movie m on channel c on day d, taking the lowest true viewership count for each channel as the worst case conversion. $(c \in C, m \in M, d \in D)$

$$A_{c,m,d}^{Avg} = \frac{1}{|\breve{T}_d|} \sum_{\breve{t} \in \breve{T}_d} \frac{TVC_{c,m,\breve{t}}^{min}}{Cost_{c,\breve{t}}^{Ads}}.$$

3.2.2 Variables

- $\eta_{m,d}$: Decision to consider movie m for day d.$(m \in M, d \in D)$
- μ_{m,w_i}: Decision to consider movie m for week w_i.$(m \in M, w_i \in W)$
- $\alpha_{c,m,d}$: Decision to consider advertising movie m on channel c on day d. $(c \in C, m \in M, d \in D)$
- γ_{c,m,w_i}: Decision to consider advertising movie m on channel c on week w_i. $(c \in C, m \in M, w_i \in W)$

3.2.3 Constraints

- Variables that are binary:

$$\eta_{m,d}, \mu_{m,w}, \alpha_{c,m,d}, \gamma_{c,m,w} \in \{0,1\}$$

- For each week w_i, there are 2 candidate days to schedule movie m:

$$\sum_{d \in w_i} \eta_{m,d} \leq 2\mu_{m,w_i}$$
$$\sum_{d \in w_i} \eta_{m,d} \geq \mu_{m,w_i}, \quad ; \forall m \in M, \forall w_i \in W.$$

- The same movie will not be played in consecutive months:

$$\sum_{w_i \in \phi_j \cup \phi_{j+1}} \mu_{m,w_i}, \quad ; 0 \leq j \leq 1, \forall m \in M$$

- Lower bound for runtime of movies on day d:

$$\sum_{m} R_m^{ads\text{-}s} \eta_{m,d} \geq 204, \quad ; \forall d \in D.$$

- Lower and Upper bound for the number of movies played on day d:

$$\sum_{m} \eta_{m,d} \leq 30 \quad ; \forall d \in D,$$
$$\sum_{m} \eta_{m,d} \geq 10 \quad ; \forall d \in D.$$

- A movie will not be selected to be played if another channel is playing that movie on day d

$$\eta_{m,d} + I_{c,m,d} \leq 1 \quad ; \forall c \in C, \forall m \in M, \forall d \in D.$$

- If movie advertised on a day d on channel c, it is advertised on its respective week w_i:

$$\gamma_{c,m,w_i} \geq \alpha_{c,m,d_i}, \quad ; \forall c \in C, \forall w_i \in W, \forall d \in w_i.$$

- If a movie is advertised in a week, the movie has to be played next week onward within 3 weeks:

$$\gamma_{c,m,w_i} \leq \sum_{w'} \mu_{m,w'} \quad ; \forall c \in C, \forall w_i \in W, w' \in \bigcup_{j=i+1}^{i+3} w_j.$$

- Upper bound of the number of advertisements shown in a day d on channel c

$$\sum_{m} \alpha_{c,m,d} \leq N_{c,d}^{ads} \quad ; \forall c \in C, \forall d \in D.$$

- The pricing for our own advertisements has to be fair compared to the rest of the market.

$$(Rev_{m,d}^{Avg} - \frac{1}{|c|} \sum_{c \in C} Cost_{c,d}^{Ads}) \cdot \eta_{m,d} \leq 50000 \quad ; \forall m \in M, \forall d \in D,$$
$$(Rev_{m,d}^{Avg} - \frac{1}{|c|} \sum_{c \in C} Cost_{c,d}^{Ads}) \cdot \eta_{m,d} \geq -50000 \quad ; \forall m \in M, \forall d \in D.$$

3.2.4 Objective Function

$$\text{Maximise} \sum_{d \in D} \sum_{m \in M} W_{m,d}^{Avg} \cdot \eta_{m,d} + \sum_{d \in D} \sum_{c \in C} \sum_{m \in M} A_{c,m,d}^{Avg} \cdot \alpha_{c,m,d}.$$

- $\sum_{m \in M} W_{m,d}^{Avg} \cdot \eta_{m,d}$ represents the value for money the candidate movies provide across the 3 months.
- $\sum_{d \in D} \sum_{c \in C} \sum_{m \in M} A_{c,m,d}^{Avg} \cdot \alpha_{c,m,d}$ represents the value for money the purchased advertisements across all competitor channels provide, across the 3 months.
- This upper layer model maximises the value for money of the candidate movies of our own channel and the purchased advertisements on the opponent channels.

3.3 Lower Layer Model

After executing the upper layer model, the lower-layer model is applied for each week, w_i, using the formulation provided below.

3.3.1 Specific Sets and Notations

- $M^A_{w_i}$: Set of candidate movies to be shown by Channel A in week w_i determined by upper layer model.

$$M^A_{w_i} = \{m \in M : \mu_{m,w_i} = 1\}.$$

- M^{ads}_{c,w_i}: Set of candidate advertisements of movies m to be shown on opponent channel c in week w_i

$$M^{ads}_{c,w_i} = \{m \in M : \gamma_{c,m,w_i} = 1\}.$$

- $N^{ads}_{c,d}$: Number of advertisements that opponent channel c shows on day d. $(c \in C, d \in D)$

3.3.2 Parameters

- $\eta_{m,d}$: Decision made by upper layer model to consider movie m for day d.$(m \in M^A_{w_i}, d \in w_i)$
- μ_{m,w_i}: Decision made by upper layer model to consider movie m for week w_i.$(m \in M^A_{w_i}, w_i \in W)$
- $\alpha_{c,m,d}$: Decision made by upper layer model to consider advertising movie m on channel c on day d. $(c \in C, m \in M^A_{w_i}, d \in w_i)$
- γ_{c,m,w_i}: Decision made by upper layer model to consider advertising movie m on channel c on week w_i. $(c \in C, m \in M^A_{w_i}, w_i \in W)$
- $R^{ads_l}_m$: Thirty minute Runtime slot with advertisement of movie m. $(m \in M^A_{w_i})$
- $S_{m,d}$: Scale Popularity of m based on demographic p. $(m \in M^A_{w_i}, p \in P)$
- $B_{p,t}$: Baseline Viewership of demographic p at time t. $(p \in P, t \in T^s_d)$
- $E^{max}_{c,t}$: Maximum true viewership count of the 10 given data for channel c at time t $(t \in T^l_d)$
- $Conv_{c,\breve{t}}$: Conversion Rate if we show advertisements at time $\breve{t}$ on channel c.$(\breve{t} \in \breve{T}_d, c \in C)$
- $U_{m,g}$: Binary matrix indicating if movie m shows genre g. $(g \in G, m \in M^A_{w_i})$

$$U_{m,g} = \begin{cases} 1 & \text{if the move } m \text{ is shows genre } g; \\ 0 & \text{otherwise.} \end{cases}$$

- K_d: Upper bound for day d, $K_d = 33 + 34d$. $(d \in w_i)$
- $TVC^{min}_{c,m,\breve{t}}$: Minimum converted true viewership if advertisement of movie m is played on day d on opponent channel c. This considers the lowest possible converted view between the 10 given data sets of true viewcount. $(c \in C, m \in M, \breve{t} \in \breve{T}_d)$

$$TVC^{min}_{c,m,d} = \min\left(\left(Conv_{c,\breve{t}}\right)\left(TVC_{c,\breve{t}}\right)\left(U_{m,g}\right)\right)$$

- $License_m$ License fee for each movie m. We note that the license fee of the movie is dependent on the movie budget and movie revenue.$(m \in M)$
- $Cost^{Ads}_{c,\breve{t}}$: Cost of buying an advertisement slot from channel c on at time $\breve{t}$. $(c \in C, \breve{t} \in \breve{T}_d)$

3.3.3 Variables

- x_{m,t_d}: Decision to schedule movie m at time t. $(m \in M^A_{w_i}, t_d \in T^l_d)$
  such that:

$$x_{m,t_d} = \begin{cases} 1 & \text{if the movie } m \text{ is scheduled at time,} t_d; \\ 0 & \text{otherwise.} \end{cases}$$

- $y_{m,d}$: Decision to play movie m on day d. $(m \in M^A_{w_i}, d \in w_i)$
  such that:

$$y_{m,d} = \begin{cases} 1 & \text{if the movie } m \text{ is played on day } d; \\ 0 & \text{otherwise.} \end{cases}$$

- $s_{m,d}$: Start time of movie m played on day d. $(m \in M^A_{w_i}, d \in w_i)$
- $e_{m,d}$: End time of movie m played on day d. $(m \in M^A_{w_i}, d \in w_i)$
- $a_{c,m,\breve{t}}$: Decision to buy advertisement slot, $\breve{t}$ of channel c to advertise a movie m.$(c \in C, m \in M^{ads}_{c,w_i}, \breve{t} \in \breve{T}_d)$ such that:

$$a_{m,c,\breve{t}} = \begin{cases} 1 & \text{if advertisement slot } \breve{t} \text{ is bought for movie } m \text{ on channel } c; \\ 0 & \text{otherwise.} \end{cases}$$

- $b_{c,m,\breve{t}}$:Converted view from channel c if we advertise movie m in time slot $\breve{t}$,

$$b_{c,m,\breve{t}} = R^{ads_l}_m \cdot TVC^{min}_{c,m,\breve{t}} \cdot a_{c,m,\breve{t}}$$

- c_{t_d} : Expected view count at time t_d $(t_d \in T^l_d)$

3.3.4 Constraints

3.3.4 Constraints

- Movie m can only be advertised on day d if the upper-layer model has considered it:

$$\sum_{t \in \breve{T}_d} a_{c,m,t} \leq \alpha_{c,m,d}, \quad (\forall c \in C,\, m \in M^{ads}_{c,w_i},\, d \in w_i)$$

- The total number of advertisement slots used is bounded by the number of ad slots available:

$$\sum_{m \in M^A_{w_i}} \sum_{t \in \breve{T}_d} a_{c,m,t} \leq N^{ads}_{c,d}, \quad (\forall c \in C,\, d \in w_i)$$

- Runtime of movie m:

$$\sum_{t_d \in T^l_d} x_{m,t_d} = R^{ads_l}_m \cdot y_{m,d}, \quad (\forall m \in M^A_{w_i})$$

- Movie m can only be played once in a week:

$$\sum_{d \in w_i} y_{m,d} \leq 1, \quad (\forall m \in M^A_{w_i})$$

- Movie m can only be played on day d if the upper-layer model has considered it:

$$y_{m,d} \leq \eta_{m,d}, \quad (\forall m \in M^A_{w_i},\, \forall d \in w_i)$$

- Only one movie can be played for every timeslot t:

$$\sum_{m \in M^A_{w_i}} x_{m,t} = 1, \quad (\forall t \in T^l)$$

- Bounds for start time of movie m:

$$s_{m,d} \leq K_d(1 - x_{m,t_d}) + t_d \cdot x_{m,t_d}, \quad (\forall m \in M^A_{w_i},\, \forall d \in w_i,\, \forall t_d \in T^l_d)$$

- Bounds for end time of movie m:

$$e_{m,d} \geq t_d \cdot x_{m,t_d}, \quad (\forall m \in M^A_{w_i},\, \forall d \in w_i,\, \forall t_d \in T^l_d)$$

- Runtime of each movie if it is played during each day:

$$e_{m,d} - s_{m,d} = (R^{ads_l}_m - 1) \cdot y_m, \quad (\forall d \in w_i,\, \forall m \in M^A_{w_i})$$

- Robust optimisation [2] by taking the minimum true viewership count as the upper bound.

$$c_{t_d} \leq \sum_{m \in M^A_{w_i}} Z_{m,t_d} \cdot x_{m,t_d}, \qquad (\forall d \in w_i, \forall t_d \in T^l_d) \tag{1}$$

$$c_{t_d} \leq 1 - \sum_{c \in C} E^{max}_{c,t_d}, \qquad (\forall d \in w_i, \forall t_d \in T^l_d) \tag{2}$$

3.3.5 Objective Function

$$\text{Maximise } 10^6 \cdot \frac{15.04}{1000} \left(\sum_{t_d \in T_d^l} 6c_{t_d} + 6 \sum_{\check{t} \in \check{T}_d} \sum_{c \in C} \sum_{m \in M} b_{c,m,\check{t}} \right) - \left(\sum_{\check{t} \in \check{T}_d} \sum_{c \in C} \sum_{m \in M} Cost_{c,\check{t}}^{Ads} \cdot a_{c,m,t_d} + \sum_{d \in w_i} \sum_{m \in M} License_m \cdot y_{m,d} \right).$$

- $\frac{15.04}{1000}$ represents the CPM (Cost per Mille). CPM calculates the cost that an advertiser will pay for every thousand impressions an ad receives. This value is taken from [6] .
- $\sum_{t_d \in T_d^l} 6c_{t_d}$ represents the total expected view count on our channel. Multiplying it by 6 ensures we calculate the view count for each 5-minute interval instead of the 30-minute interval.
- $6 \sum_{\check{t} \in \check{T}_d} \sum_{c \in C} \sum_{m \in M} b_{c,m,\check{t}}$ represents the total converted view count on our channel. Multiplying it by 6 ensures we calculate the view count for each 5-minute interval instead of the 30-minute interval.
- $\sum_{\check{t} \in \check{T}_d} \sum_{c \in C} \sum_{m \in M} Cost_{c,\check{t}}^{Ads} \cdot a_{c,m,t_d}$ represents the cost of purchasing advertisements on other channels.
- $\sum_{d \in w_i} \sum_{m \in M} License_m \cdot y_{m,d}$ represents the license fees of each movie incurred from showing them on Channel A.

4 Results and Analysis

Channel	Base True View Count ($\times 10^9$)	After Conversion				
		Converted View ($\times 10^9$)	Total View ($\times 10^9$)	Ad Cost/Revenue (£$\times 10^6$)	License Fee (£$\times 10^6$)	Profit (£$\times 10^6$)
Channel A	1.23	+1.12	2.36	−10.61	−22.20	+2.62
Channel 0	0.82	−0.10	0.72	+1.08	−208.29	−196.40
Channel 1	1.58	−0.39	1.19	+3.86	−203.83	−182.13
Channel 2	1.91	−0.63	1.27	+5.67	−201.74	−176.92

Table 1: Viewcounts with Converted Views & Revenue and Costs

4.1 Analysis on Viewership and Profit

The data presented in Table 1 describes the baseline viewer counts of various channels before and after advertisement-driven conversion. It should be noted that the advertisement slots of Channel A are marketed on the open market, leading to the assumption within these results that conversions of viewers from other channels are exclusively attributed to Channel A. Given that Channel A purchases advertisement slots from other channels, it incurs costs, whereas the selling channels realize profits from these transactions.

- Channel A's base view count of 1.23 billion is 11.19% below the market average. However, its spending on license fees, at -22.20 million, is 86.04% less than the average for the market. **This substantial difference highlights that Channel A achieves a competitive number of views while spending considerably less on license fees.**
- Approximately 47.46% of Channel A's viewership is attributed to conversion from advertisements. **This indicates that nearly half of the total views after the advertising efforts are a result of the conversions.**
- The advertising cost per converted view for Channel A is approximately £0.0095, which represents the expenditure required to gain each additional viewer through advertisement-driven conversion. The license fee per total view, which accounts for all viewers after conversion (both base and converted), is approximately £0.0094. Comparing these values shows that the advertising cost per converted view and the license fee per total view are almost equivalent. **This suggests that, in the long run, the overall cost-effectiveness of advertising and licensing strategies is similar.**

4.2 Time Series Analysis

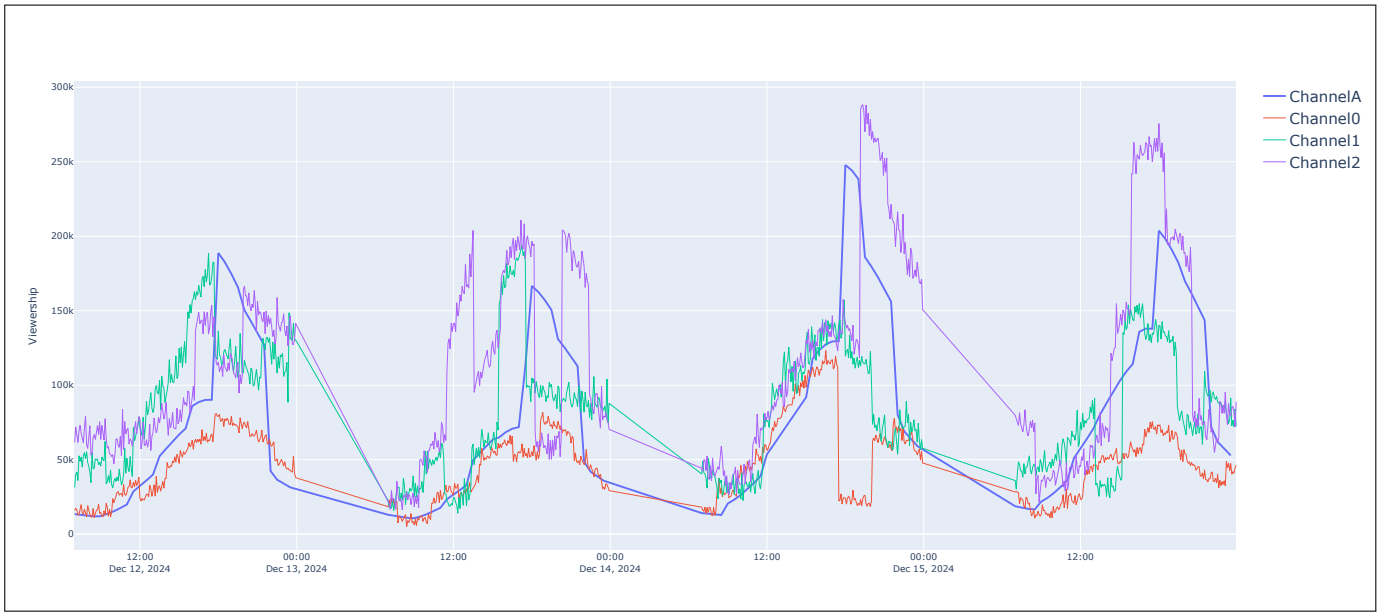


Figure 4: True View Count Comparison among 4 channels

The Figure 4 illustrates the example of how true viewership changes over time for all Channels during 12th December to 15th December.

- The market gap, in this context, refers to periods when other channels are not investing in maximizing their peak time, yet these periods remain optimal for profitability.
- While Channel A may not achieve the highest total true view count, it strategically dominates the most critical timeframe—prime-time (6 PM to 4 PM)—over a consistent three-month period. This demonstrates the model's capability to not only identify the most profitable time slots but also mitigate the risks associated with high license fees.
- For weekly performance, the model adopts a similar approach by strategically avoiding certain peak weeks. Although this results in some profit loss during those weeks, Channel A remains profitable overall, underscoring the model's ability to balance short-term sacrifices for long-term gains.
- Regardless of the given data, there is a significant market gap on weekends, as Channels 1 and 2, which account for 65% of the market, tend to focus their competition on the Monday and Saturday periods. Consequently, our model heuristically allocates more resources to Saturday and Sunday, a strategy that proves to be both practical and effective.

4.3 Advertisement Cost Analysis

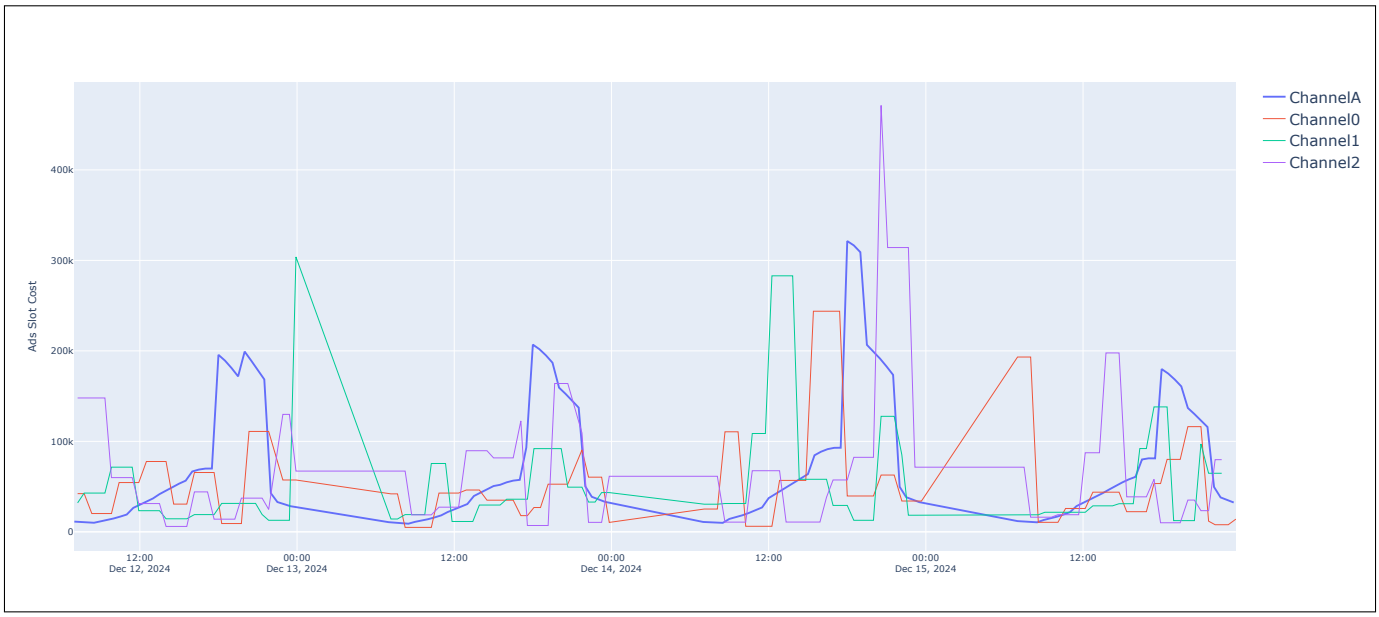


Figure 5: Advertisement Slot Price

Figure 5 shows the price of the advertisement slots of our own channel from December 12 to December 15.

- Referring to Figure 4, there is a notable correlation between advertisement slot pricing and the true view count. This relationship is driven by the pricing model, which adjusts advertisement rates based on viewership. This strong correlation is significant as it implies that the value of advertisement slots increases proportionally with viewership, providing greater opportunities for converting viewers to other channels.
- Additionally, it can be observed that the advertisement prices are closely aligned with those of the broader market. This indicates that the pricing strategy remains competitive and fair, which is essential to ensure that the market continues favourably considering purchasing advertisement slots on Channel A.

4.4 Revenue Analysis

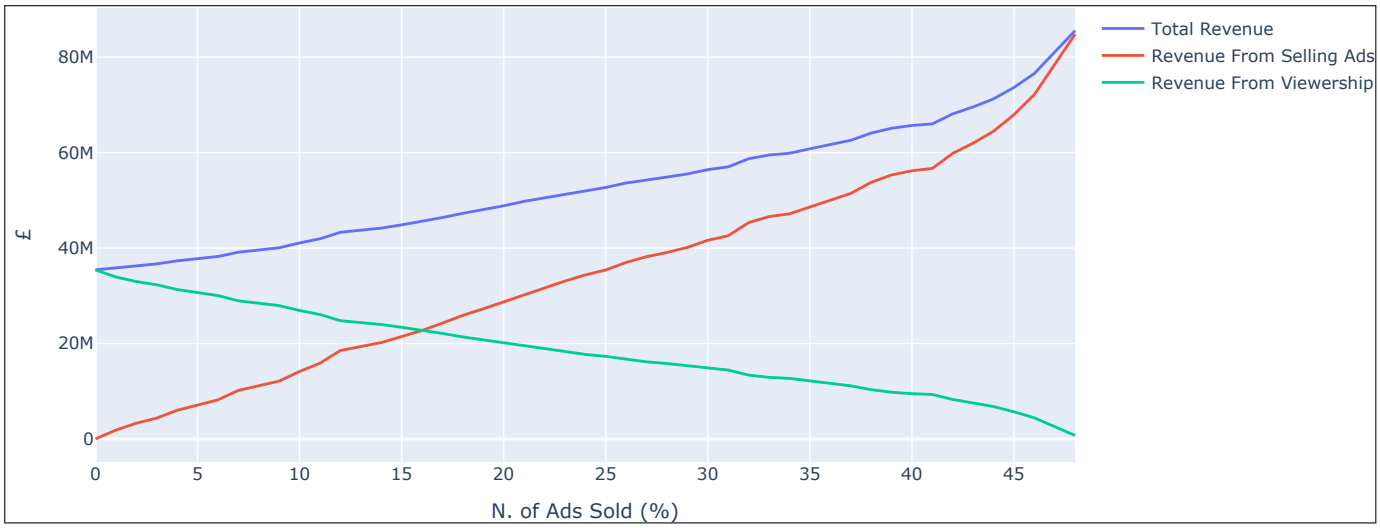


Figure 6: Revenue from Selling Ads vs Viewership

Figure 6 shows the total revenue from selling advertisements and viewership, and the effects of varying the number of ads sold on them.

- To evaluate the effectiveness of the model, we consider scenarios in which other channels purchase advertisement slots from Channel A. It is assumed that these slots are purchased based on a prioritized ranking derived from the ratio of Channel A's viewership to associated costs. In a worst-case scenario, all viewership for a given slot

is converted to other channels. Consequently, as the number of sold advertisement slots increases, there is a decrease in revenue from viewership at a rate of 1:5, indicating that for every five viewers gained, one unit of viewership revenue is lost. Meanwhile, revenue from selling advertisements increases at a rate of 4:3, implying that for every three advertisement slots sold, revenue increases by four units. This dynamic highlights a trade-off between maintaining viewership and generating income through advertisement sales.

- For cases where the number of sold advertisements exceeds 15%, the total revenue will primarily rely on revenue from sold advertisements rather than viewership. Furthermore, in the worst-case scenario, selling 50% of the advertisement slots would result in the complete loss of viewership.

4.5 Demographic Analysis

Based on our analysis, we observe that the model targets the underserved demographic, in particular the retirees.

- Based on the provided data, there is a significant market gap within one demographic group: retirees. Retirees account for only 17% of the total True View count in other channels. Consequently, the model suggests placing greater emphasis on this demographic, which directly influences the selection of movies for Channel A.
- As illustrated in Fig.7, Channel A is notably recommended to feature a higher proportion of Drama films. This is because movies in this genre tend to resonate well with the retiree demographic. While retirees' scaled popularity averages around 0.36, the Drama genre exceeds this average, scoring above 0.4. Conversely, there is a noticeable decline in the recommendation for Action and Adventure films, as these genres generally appeal less to retirees.
- The movie eras have been categorized into three main groups: old (1950s–1970s), medium (1980s–2000s), and new (2010s–2020s). Typically, newer movies tend to be more popular overall. However, as shown in Fig.8, the model specifically recommends selecting films from the old category. These older films offer high cost-efficiency and demonstrate strong popularity among retirees, with their popularity factor exceeding 36%.

5 Conclusion

Channel A employs a comprehensive strategy to optimize revenue and audience engagement through:

- **Competitive Advertisement Pricing**: From Figure 5, advertising slots are offered at competitive rates, dynamically adjusted based on viewership, ensuring value for advertisers and market competitiveness.
- **Cost-Effective Movie Selection**: From Table 1, movies are chosen based on a ratio of popularity to license fee, ensuring high-value content with minimal cost.
- **Peak Viewership Optimization**: From Figure 4, Time series analysis is used to schedule content during periods of low competitor activity, capturing peak viewership opportunities.
- **Targeting Underserved Demographics**: From Section 4.5, content is curated to appeal to underserved audiences, in particular the retirees, by featuring older movies and scheduling at suitable times.

These strategies collectively enable Channel A to:

- Maintain a competitive edge in the market.
- Attract valuable audiences and provide value-driven content.
- Offer an attractive platform for advertisers to maximize their reach.

With the current model, a total budget of £32.81 million is required, comprising £22.2 million allocated for licensing fees and £10.61 million designated for the advertisement budget. This current model projects profit at £2.62 million, assuming that other channels do not buy our advertisement slots. From 6, we note that if they do decide to buy our advertisement sots, our revenue will increase despite the loss in viewership.

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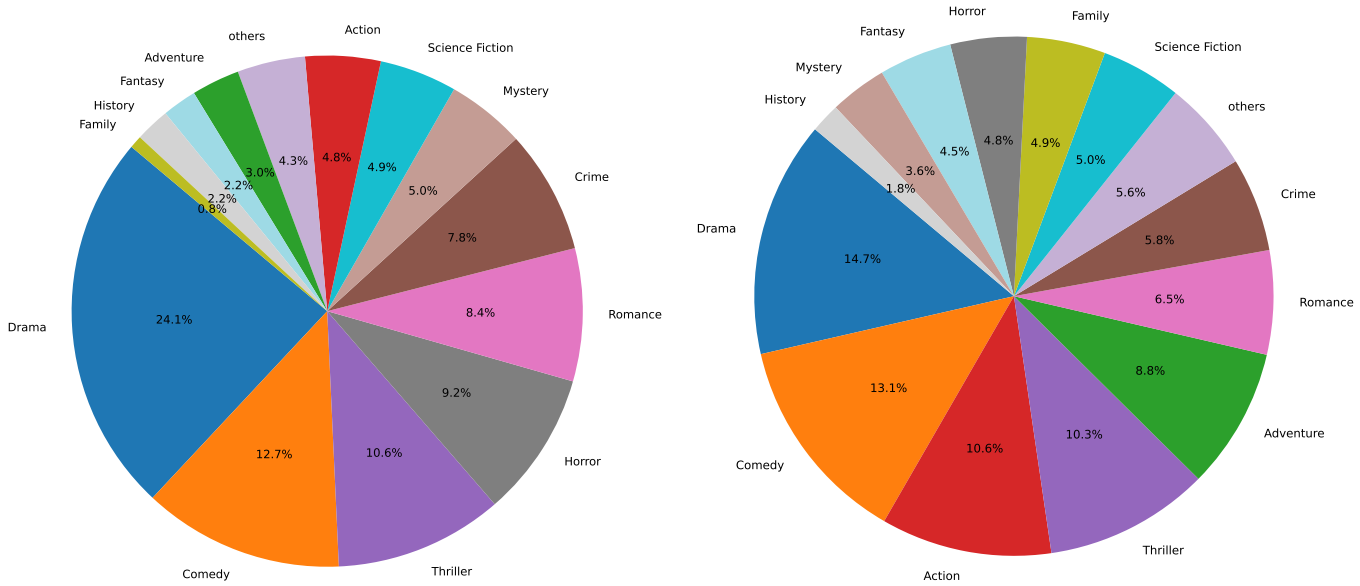
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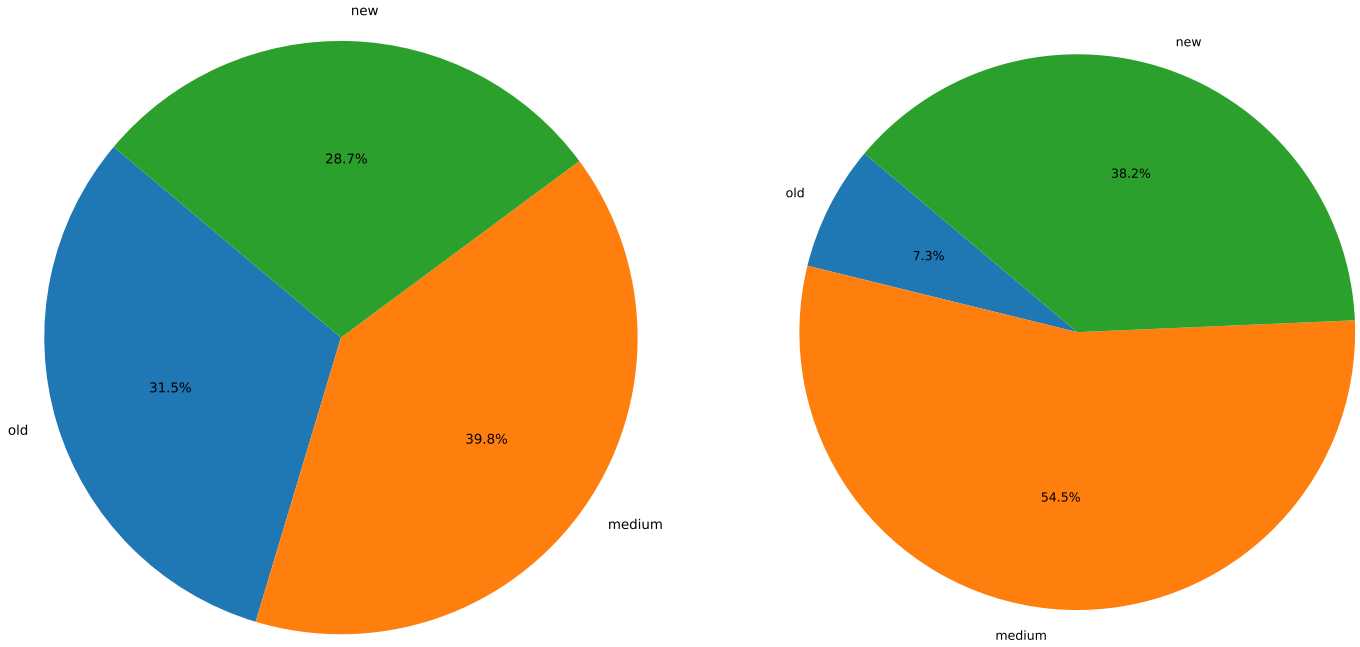
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6 Annex



(a) (b)

Figure 7: (a) ; (b)



(a) (b)

Figure 8: (a) ; (b)